

Deriving Trigonometry Values with a Stick

By Nico P.

You are a ship's carpenter on a sailing ship that broke its mast. Your ship stops at an island to find a tree to replace it. The tree must be between 70-75 feet tall. You don't have time to cut down trees that are not the right size. How would you measure the heights of the trees?

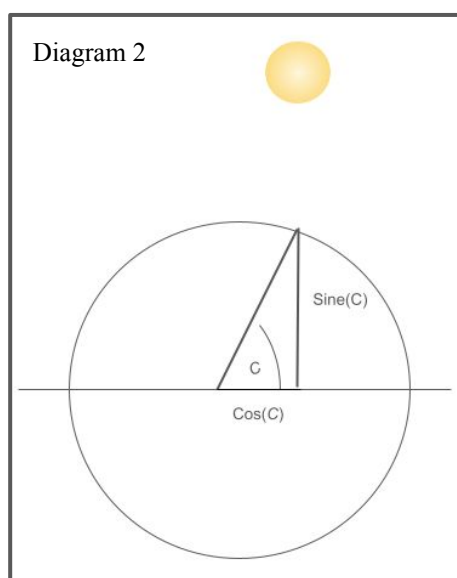
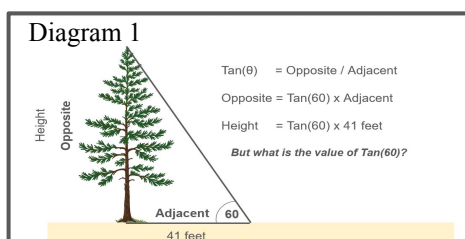
Trigonometry can help you find the right tree. Using a ruler and the ship's sextant, a tool which measures angles, there's an easy way to find the heights of the trees:

To measure the height of a tree, walk back from the tree and measure the distance you walked, then measure the angle to the top of the tree. Using these measurements and simple trigonometry, we can find the tree's height.

Let's suppose you walked 41 feet away from the tree, and the angle to the treetop was 60 degrees. You can use the tangent ratio (opposite / adjacent) to solve for the height of the tree. In this case, "opposite" is the height of the tree, and "adjacent" is the distance from the tree. We can see that $\tan(60^\circ) = \text{height}/41$.

The problem is you don't remember the value of $\tan(60^\circ)$ and you don't have a table of trigonometry values. How would you calculate $\tan(60^\circ)$?

One method uses a stick and its shadow to create a simple unit circle.



Cosine

First we will use the stick to demonstrate Cosine (Adjacent/Hypotenuse) for the angle created by the ground and the stick. (Diagram 2).

The shadow's length is the "adjacent" in our calculations. Cosine = Adjacent/Hypotenuse. Since our Hypotenuse is 1, then the shadow always represents the value of the cosine of that angle.

You can visualize cosine by holding a stick at a 0 degree angle to the sun (**90 degrees to the flat ground**), so that it casts no shadow ($\cos(C) = 0$), while a horizontal stick at 90 degrees to the sun (**0 degrees to the flat ground**) casts a shadow equal to its length ($\cos(C) = 1$).

Remember, we are taking the cosine of the angle created by the ground

and the stick. So Cosine = Adjacent/Hypotenuse. And since the hypotenuse = 1, $\cos(C)$ is the value for Adjacent. So Cosine is always the length of the shadow.

Sine

Sine = Opposite/Hypotenuse. Using the same method, and the same angle C, sine is the distance from the top of the stick to the end of the shadow ("Opposite"), divided by the stick's length ("Hypotenuse") (Diagram 2).

Tangent

Tangent = Opposite/Adjacent. To calculate tangent of angle C ($\tan(C)$) we take the distance from the top of the stick to the end of the shadow ("Opposite") and divide it by the length of the shadow ("Adjacent"). Note that Tangent is also the value of sine divided by cosine (Diagram 2).

In a unit circle, we see that "the height of the stick" is the value, and "the length of the shadow" is the x value. However, x and y can be negative, while lengths cannot. This can be resolved for x by drawing a line that passes through the stick, and counting shadows that fall on one side as negative, and counting ones that fall on the other side as positive.

When you measure the shadow of the stick at 60 degrees you get a length of 0.5, and you measure the height and you get ~ 0.886 . Since Tangent = Opposite/Adjacent, $0.886/0.5 \approx 1.73$. $\tan(60^\circ) = 1.73$

Returning to the problem, we get 41 feet $\times 1.73 = 70.93$ feet. The tree is the right height!

Spring Math Crossword

By Kevin C.

ONE DOWN

Consider a polar rose given by the equation $r = \frac{8}{3} \cos(8\theta)$. Given that the area of one "petal" can be expressed in the form $\frac{m\pi}{n}$ where m and n are relatively prime positive integers. Find $m + n$.

THREE ACROSS

A hare is frolicking in the middle of a grass field which is a 3-4-5 right triangle. Given that n is the shortest distance the hare can travel to reach each of the corners of the field at least once, find the integer closest to the maximum value of n .

ONE ACROSS

Consider the size of a bunny population modeled by the logistic growth curve function $P(x) = \frac{15}{(1 + Ae^x)^{1/v}}$, where $x \in \mathbb{R}$. What is the carrying capacity (maximum population size) of this bunny population?

TWO DOWN

The variable L represents the length (in inches) of a randomly selected snake from the grass field. Given that $L \sim N(30.5, 1.734^2)$, and the probability that L is odd when rounded to the nearest integer is approximately $n\%$, find n .

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